

Chapter 7 – Advanced Nmap Scanning & Wireshark Packet Analysis

Introduction to Advanced Nmap Scanning

In the previous lecture, basic scanning was learned.

In this lecture, advanced scanning techniques were introduced.

These techniques are used for:

- ✓ Firewall evasion
 - ✓ Stealth scanning
 - ✓ IDS/IPS bypass attempts
 - ✓ Advanced reconnaissance
 - ✓ Packet-level analysis
-

Understanding TCP Flags

Before understanding advanced scans, TCP flags are very important.

TCP communication uses different flags to control connections.

Common TCP Flags

Flag	Full Form	Purpose
SYN	Synchronize	Start connection
ACK	Acknowledgement	Confirm received packet
FIN	Finish	Close connection
RST	Reset	Terminate connection immediately
PSH	Push	Push data immediately
URG	Urgent	Urgent data

Flag	Full Form	Purpose
NULL	No flag	Used in special scans



Daily Life Example

Imagine a phone call:

- SYN → "Can we talk?"
- SYN-ACK → "Yes, I'm available."
- ACK → "Okay, connection established."
- FIN → "Conversation finished."
- RST → "Connection rejected."



SYN Scan (Stealth Scan)



What is SYN Scan?

SYN Scan is one of the most popular Nmap scanning techniques.

It is also called:



Half-Open Scan

because the TCP handshake is never fully completed.



Command

`nmap -sS <target>`



Example

`nmap -sS 192.168.1.10`

How SYN Scan Works

- 1 Nmap sends SYN packet to target port
 - 2 If port is open → Target replies SYN-ACK
 - 3 Nmap immediately sends RST packet
 - 4 Connection terminates before full handshake
-

If Port is Closed

Target replies:

RST

This means:

✗ Port is closed

If Firewall Exists

Sometimes no reply comes.

Nmap shows:

filtered

This usually means:

🔥 Firewall is blocking packets.

Why SYN Scan is Important?

- ✓ Fast
 - ✓ Stealthy
 - ✓ Less logging
 - ✓ Common in penetration testing
-

Real-World Analogy

Imagine slightly knocking on a door and leaving before the person fully opens it.

That is how SYN scan behaves.

Xmas Scan

What is Xmas Scan?

Xmas Scan sends packets with:

- ☒ FIN
- ☒ URG
- ☒ PSH

flags enabled together.

The packet looks “lit up like a Christmas tree.”

Hence the name:

Xmas Scan

☒ Command

```
nmap -sX <target>
```

Example

```
nmap -sX 192.168.1.10
```

How Xmas Scan Works

Port State	Target Response
------------	-----------------

Open Port	No response
-----------	-------------

Port State Target Response

Closed Port RST packet

Important Limitation

⚠ Works mainly on Unix/Linux systems.

Windows systems generally do not follow RFC behavior correctly and may respond differently.

Why Use Xmas Scan?

Used for:

- ✓ Firewall evasion
 - ✓ IDS bypass attempts
 - ✓ Stealth scanning
-

NULL Scan

What is NULL Scan?

NULL Scan sends TCP packets with:

 No Flags Set

✓ Command

`nmap -sN <target>`

Example

`nmap -sN 192.168.1.10`

How NULL Scan Works

Port State	Response
------------	----------

Open Port	No response
-----------	-------------

Closed Port	RST
-------------	-----

Windows Limitation

Windows usually replies with RST for both open and closed ports.

Therefore:

 NULL scans are unreliable on Windows.

FIN Scan

What is FIN Scan?

FIN Scan sends packets with only:

 FIN Flag

enabled.

Command

```
nmap -sF <target>
```

Example

```
nmap -sF 192.168.1.10
```

How FIN Scan Works

Port State Response

Open Port No response

Closed Port RST

Purpose of FIN Scan

- ✓ Helps bypass simple firewalls
 - ✓ Avoids traditional SYN detection
 - ✓ Useful for stealth scanning
-

Daily Life Example

Instead of ringing the bell normally (SYN), you quietly test the door handle (FIN).

Timing Templates in Nmap

Purpose

Timing templates control:

Scan Speed vs Stealth

✓ Command

nmap -T4 <target>

Timing Levels

Level Name	Purpose
T0	Paranoid Very slow, stealthy
T1	Sneaky Slow stealth
T2	Polite Less bandwidth usage
T3	Normal Default
T4	Aggressive Faster scanning
T5	Insane Extremely fast

Important Concept

Fast Scan	Stealth Scan
Quick results	Harder detection
More noise	Slower

Example

```
nmap -sS -T4 192.168.1.10
```

MAC Address Spoofing

What is MAC Spoofing?

MAC Spoofing changes your device's MAC address to hide identity.

Why Use It?

Used for:

- ✓ Identity masking
 - ✓ Bypass MAC filtering
 - ✓ Mimic another vendor/device
-

✓ Command

```
nmap --spoof-mac <value> <target>
```

Example – Specific MAC

```
nmap --spoof-mac 00:11:22:33:44:55 192.168.1.10
```

Example – Vendor MAC

```
nmap --spoof-mac Apple 192.168.1.10
```

Example – Random MAC

```
nmap --spoof-mac random 192.168.1.10
```

Daily Life Example

Like wearing a fake ID card to hide your real identity.

Decoy Scan

What is Decoy Scan?

Decoy scan uses fake IP addresses to confuse logs and IDS systems.

✓ Command

```
nmap -D decoy1,decoy2,ME <target>
```

✖ Example

```
nmap -D 10.0.0.1,10.0.0.2,ME 192.168.1.10
```

📘 Meaning of ME

ME represents:

💻 Your Real IP Address

🎲 Random Decoys

```
nmap -D RND:5 192.168.1.10
```

This generates:

🎭 5 Random Fake IPs

🎯 Purpose

- ✓ Hide real source
 - ✓ Confuse IDS/IPS
 - ✓ Increase anonymity
-

Summary Table

Scan Type	Command	Purpose	Works Best On
SYN Scan	<code>nmap -sS</code>	Stealth scanning	All systems
FIN Scan	<code>nmap -sF</code>	Firewall evasion	Unix/Linux
NULL Scan	<code>nmap -sN</code>	Stealth	Unix/Linux
Xmas Scan	<code>nmap -sX</code>	Firewall evasion	Unix/Linux
Decoy Scan	<code>nmap -D</code>	Hide identity	All systems



Wireshark Introduction



What is Wireshark?

Wireshark is a:



Network Packet Analyzer

It captures and analyzes network traffic.



What Wireshark Can Show

- ✓ Source IP
- ✓ Destination IP
- ✓ MAC addresses
- ✓ TCP/UDP packets
- ✓ Protocols
- ✓ Packet data
- ✓ Ports
- ✓ Network conversations

Daily Life Example

Wireshark works like:

CCTV Camera for Network Traffic

It records everything moving inside the network.

Opening Wireshark in Kali Linux

Command

wireshark

Selecting Interface

After opening Wireshark:

You will see interfaces like:

eth0
wlan0
lo

Common Interfaces

Interface Meaning

eth0 Ethernet

wlan0 Wi-Fi

lo Loopback

Packet Capture

Click on active interface (example: eth0).

You will see live traffic graphs.

Then Wireshark starts capturing packets.

Important Toolbar Buttons

Button	Purpose
Blue Shark Fin	Start capture
Red Square	Stop capture
Restart Icon	Restart capture
Save Icon	Save packets
Search Icon	Search traffic

Packet Capture File Extension

Wireshark saves files as:

.pcap

Packet Structure in Wireshark

When clicking a packet, you can see layers:

Ethernet Layer

Shows:

- ☒ Source MAC
 - ☒ Destination MAC
-

IP Layer

Shows:

- ☒ Source IP
- ☒ Destination IP

TCP/UDP Layer

Shows:

- ☒ Source Port
- ☒ Destination Port
- ☒ Flags

Data Layer

Shows actual packet payload/data.

Sometimes encrypted.

Sometimes readable.

Searching Packets in Wireshark

☒ Filter Example

ip.addr == 192.168.1.16

Purpose

Shows only packets related to that IP address.



Packet Information Columns

Wireshark displays:

Field	Meaning
No.	Packet number
Time	Timestamp
Source	Sender IP
Destination	Receiver IP
Protocol	TCP/UDP/HTTP etc.
Length	Packet size
Info	Additional details



HTTP Downgrade Attack Concept



Educational & Defensive Understanding Only

This concept was discussed theoretically for understanding network risks.



Basic Idea

In insecure networks, attackers may attempt:

- Traffic interception
- HTTPS downgrade attempts
- Man-in-the-Middle positioning

to view unencrypted HTTP traffic.

Important Networking Concepts Mentioned

To perform network interception theoretically, attacker would need:

- ✓ Access to same network
- ✓ Router/gateway information
- ✓ ARP table details
- ✓ MAC addresses

Why HTTPS Matters

HTTPS encrypts communication.

HTTP does not.

Daily Life Example

HTTP

HTTPS

Open postcard Locked envelope

ARP Table Concept

ARP table maps:

 IP Address ↔ MAC Address

Why ARP is Important?

Devices need MAC addresses to communicate inside local networks.

Why Ethical Hackers Learn Packet Analysis

Packet analysis helps in:

- ✓ Troubleshooting
 - ✓ Malware analysis
 - ✓ Detecting attacks
 - ✓ Understanding protocols
 - ✓ Network forensics
 - ✓ Traffic monitoring
-

Legal & Ethical Warning

Wireshark and Nmap are powerful tools.

Never capture or analyze traffic without permission.

Unauthorized sniffing may violate:

- Privacy laws
- Cybercrime laws
- Organizational policies

Always practice inside:

- ✓ Personal labs
 - ✓ Virtual machines
 - ✓ Authorized environments
-